

Seismic attributes in reservoir characterization: an overview

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Abstract

The present study focuses on a literature review using seismic attributes in the characterization of hydrocarbon reservoirs. This work consists of examining the contribution of seismic attributes in the characterization of reservoirs. With this, it is necessary to identify and list the attributes involved in analyzing the reservoirs and their degree of involvement in the characterization of the hydrocarbon accumulation zones. The approach used is to classify attributes according to their levels of intervention as the attributes that circumscribe hydrocarbon accumulation zones such as instantaneous frequencies. Thus, the attributes which make it possible to analyze the seismic facies must be enumerated to determine the type of hydrocarbons. The aim of this work is not only to review the literature on the use of seismic attributes in the characterization of reservoirs but also to expose the limitations of these attributes in the study of reservoirs to search for the most adapted attributes that have never been used in reservoir analysis. As a result, several seismic attributes used in reservoir characterization will be evaluated while demonstrating their strengths and limitations to consider either the combination of certain attributes or the experimentation of new attributes in deepening the knowledge on the study of reservoirs. This work will contribute to a more sophisticated study in this domain.

Keywords Seismic attributes · Characterization · Reservoir · Hydrocarbon

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Introduction

In general, there are two types of reservoir characterizations, namely static characterization and dynamic characterization. The static characterization is to determine the properties of reservoir rocks in a given area. This is an important step to understand the spatial distribution of reservoir layer parameters and their environments. Dynamic characterization involves performing an evolutionary analysis of reservoir properties to evaluate existing fluids over time. This approach is mainly used to plan and control production. This article focuses mainly on static characterization, which will review the work done on the use of seismic attributes in the analysis of reservoir properties. Thus, seismic attributes are special measures of kinematic, dynamic, geometric, or statistical characteristics obtained from seismic data and which contribute to structural, stratigraphic, and petrophysical interpretation (Djeddi 2016). In recent years, several attributes have been used in reservoir characterization to improve the understanding of reservoir characteristics. The introduction of a multitude of attributes has revolutionized oil exploration and has extended the application of the

attributes in the reservoir analysis. It begins with the location of trap structures, then follows the determination of accumulation zone parameters, and ends up with the detection of hydrocarbon indicators.

Reminders of reservoir characteristics

Reservoir characterization consists of analyzing the different reservoir parameters such as porosity, permeability, water saturation, reservoir lithology, and sometimes its size. Reservoirs are defined in this work as rocks with interconnected voids that promote the circulation and accumulation of fluids. If we find the hydrocarbons (oil and/or gas) are found within among these fluids, reservoirs are called hydrocarbons. The complicity of reservoirs sometimes makes it difficult to better understand these reservoir rocks to the spatial and even temporal variation of the criteria which conditions the existence of reservoir rocks. Nevertheless, reservoirs can be defined according to the petrophysical and geological characteristics.

Petrophysical characteristics

Petrophysics is the study of the physical and chemical characteristics of rocks. According to Cossé (1988), a rock constitutes a reservoir if and only if it has a storage capacity (porosity) and that fluids can circulate (permeability). The petrophysical characteristics of the reservoirs are closely dependent on the rock-fluid relations and in particular the interfaces between rocks and their contents (fluids). There are two types of petrophysical properties, namely conventional petrophysical properties and those derived from rock mechanics (Neau 2009).

Conventional petrophysical properties

It is necessary to take a look at the elements that characterize the reservoirs notably the porosity, permeability, fluid saturation, and volume of clay.

The porosity (Φ) of a rock is its property to have voids, pores, or cracks. It is expressed quantitatively by the percentage of the pore volume in relation to the total volume of the rock. Consider a rock sample with its apparent volume or total volume V_T consisting of a solid volume V_S and a pore volume V_P . The porosity Φ is presented in the equation: $\Phi = \frac{V_{\text{Void-space}}}{V_{\text{Total}}}$, expressed in %.

Reservoir specialists are interested in porosity allowing the circulation of fluids in the pores. Φ_e is the effective porosity which corresponds to the pores connected to each other and the outside. Thus, we define the total porosity Φ_t corresponding to all the pores, whether or not related to each other, and the residual porosity Φ_r which considers only isolated pores.

The pore structure varies depending on the type of reservoir rock because the pore system of the carbonate reservoir rock is different and irregular. Thus, the pore structure has an impact on the elastic wave velocities. Sun (2004) demonstrates that by considering the resolution of Zoeppritz's exact equations and elastic finite element modeling, it is possible to detect variations in pore structure in reservoir rocks from seismic data via an amplitude variation analysis with offset (AVO).

Permeability is a measure of the capacity for fluid movement in porous or fractured media (Rubino et al. 2012). If the fluid is homogeneous and has no significant action on the surrounding medium, the permeability is said to be absolute. The unit of measurement of the permeability is the darcy, and in most cases, the millidarcy is used (with $1 \text{ mD} = 10^{-15} \text{ m}^2$). The permeability is good if its value is between 200 and 500 mD.

Saturation can be defined as impregnation of interstitial voids of a porous or cracked medium by infiltration of fluid (oil, gas, or water). Thus, the saturation of oil, water, and gas is defined as the ratio of each of the volume of the fluids to the total volume and they are denoted as S_W , S_O and S_G for the water saturation, oil, and gas respectively (1).

$$S_W = \frac{V_W}{V_P}, \quad S_O = \frac{V_O}{V_P}, \quad S_G = \frac{V_G}{V_P} \quad (1)$$

where V_P is the pore volume, V_W is the volume of water, V_O is the volume of oil, and V_G is the volume of gas. In the petroleum domain, knowledge of the oil and gas volumes requires knowledge of saturation at all points or at least a satisfactory approach (Cossé 1988).

Volume of clay The volume of clay is very important for the study of reservoir rocks. So, we consider the case of simple clay formations that are formed from a matrix rock sandstone, limestone, or dolomite, mixed with clay. They contain interstitial fluids such as water, oil, or gas. The distribution of the clay usually gives dispersed, laminated, and structural clays. The clay volume V_{CL} comprises bound water ϕ_B but sometimes the clay index (I_{CL} or I_{sh}) is used instead of clay volume because of its relation to the effect of the clay only to its volume. In fact, V_{sh} consists of clay and the impurities associated with it that are silt, limestone, and anhydrite while V_{CL} is considered as the volume of the pure clay. It is this pure clay that intervenes during the interpretation.

Petrophysical properties derived from rock mechanics

The indirect or derived petrophysical properties from mechanical rocks are basically related to elastic modules (incompressibility, shear modulus, Young's modulus, Poisson's ratio). The reservoir is assumed to be a porous isotropic and linearly elastic medium. Thus, the velocities of the

P and S waves are entirely characterized by three parameters, namely the isostatic modulus of elasticity (bulk modulus), the shear modulus (shear modulus), and bulk density which is in grams per cubic centimeter. The rock contains a solid component and porous component saturated with different fluids that represent a continuous real medium with average elastic properties.

Geological characteristics

Geological characteristics are important in the determination of layers that constitute accumulation zones. Thus, reservoir rocks can be grouped into two main types, namely detrital reservoirs and carbonate reservoirs.

Detrital reservoirs (also known as sandstone reservoirs)

Detrital reservoirs are generally composed of stable elements, having a fairly homogeneous porosity and permeability, of intergranular type, controlled by the grain size and the degree of cementation. The relationships between porosity and permeability are generally good and decrease with landfill that are regular and function with temperature.

Carbonate reservoirs

Carbonated reservoirs are characterized by a very large heterogeneity of porosity and permeability. They are often massive, large pore, and fissured in the case of reefs, finely stratified and laminated in the reef and shoal deposits. They consist mainly of dolomites and limestones.

Characterization of reservoirs by seismic attributes

The characterization of the reservoirs by the analysis of the seismic attributes (Oyeyemi and Aizebeokhai 2015) enables to build from these attributes so as to make a precise study of the reservoirs to determine the different models of the reservoirs which are precisely geological, stratigraphic, and petrophysical models. The seismic attributes make it possible to estimate the permeability, porosity, hydrocarbon indicators, fluid movement, thickness, lithology, and the monitoring of hydrocarbon production. This study does not enumerate exhaustively all the attributes used in reservoir characterization but highlights some relevant seismic attributes best suited for reservoir location, property analysis, and hydrocarbon zone detection.

Location of reservoir

Nowadays, there are attributes that can circumscribe areas of hydrocarbon accumulations. This localization takes place thanks to the structures that define the delimitation and the location of the reservoirs because it involves identifying trap structures. There are summarily three types of traps: structural traps (faults, anticlines, and synclines); stratigraphic traps (sandstone body, carbonate reef, and permeability bevel); and mixed traps (salt dome). Several seismic attributes such as similarity, thinned fault likelihood, semblance, fault enhancement filter, ridge enhancement filter, curvature, variance, azimuthal AVO, inversion to anisotropic parameters, fault extraction, instantaneous amplitude, sweetness attribute, coherence, gradients, and dips are used to determine the structures for locating the reservoirs. But the curvature, coherence, gradient, and dip are mainly dedicated to the determination of channel, faults, anticlines, synclines, and salt dome. To do this, it would be necessary to highlight the most important attributes in the detection of traps such as coherence, variance, amplitude gradients, dip azimuth measurements, and curvature. Coherence attribute can be defined as a measurement of the lateral continuity of the seismic signal (Backé et al. 2012). The identification of faults by coherence (Xia et al. 2015) is undoubtedly a topic that interests specialists in oil exploration based on publications in this domain (Fig. 1). But the first publications (Bahorich and Farmer 1995; Marfurt et al. 1998; Garsztenkom and Marfurt 1999) played a decisive role in the use of this method, because it was in their work to highlight a cross-correlation of the seismic traces, then to conduct a study of multitrace and finally to implement the algorithm of similarity obtained thanks to the eigenvalues of a matrix of covariance on a window of seismic data. For example, several studies (Roberts 2001; Sigismondi and Soldo 2003; Chopra and Marfurt 2007a) were conducted to improve fault detection. With coherence, the location of reservoirs becomes easy by providing for the determination of geological structures (Rezvandehy et al. 2011; Bahorich and Farmer 1995). Coherence attribute is the method used in calculating the similarity between seismic traces and is also the best determinant of anticline structures, faults, deltas, river channels, dikes, and other stratigraphic anomalies (Rezvandehy et al. 2011). The identification of faults and fracture zones by the coherence attribute was first illustrated by the works of Bahorich and Farmer (1995) on the cross-correlation of seismic traces, then by a study on a multitrace semblance of Marfurt et al. (1998) and finally by the analysis of Garsztenkom and Marfurt (1999) based on the similarity algorithm obtained thanks to the eigenvalues of a covariance matrix on a window of the seismic data. Thus, the detection of faults by the coherence attribute has improved considerably thanks to the research conducted in recent years (Randen et al. 2001; Pedersen et al. 2002; Aqrawi and Boe 2011; Dorn et al. 2012).

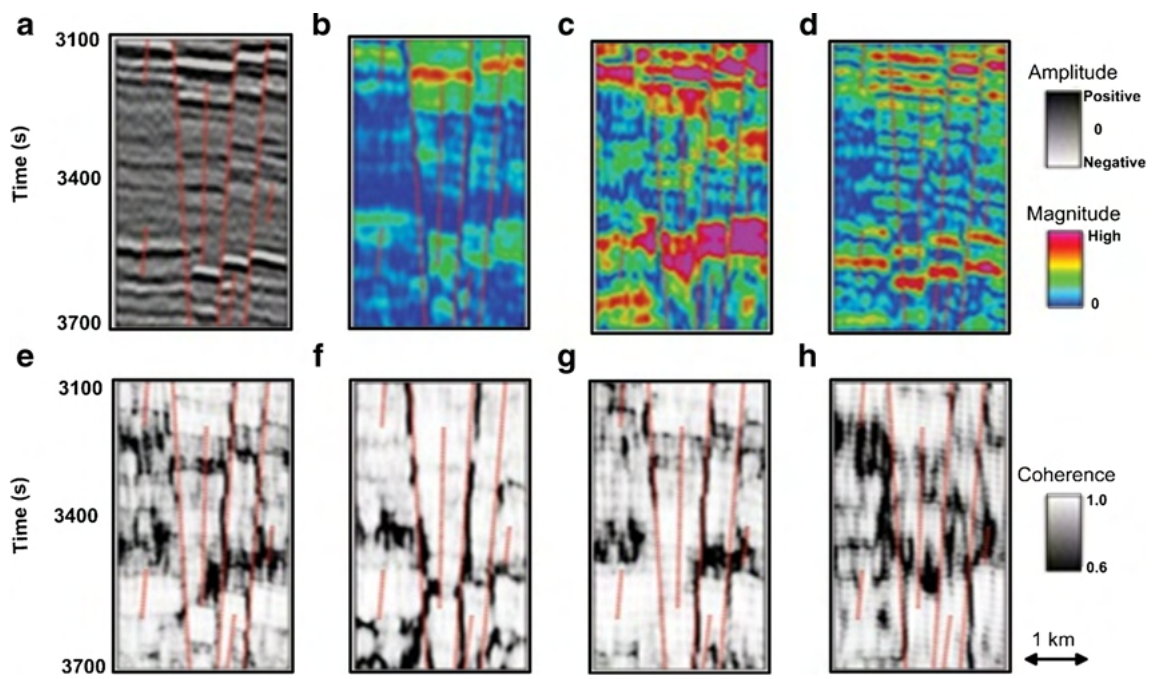


Fig. 1 Fault detection by complex spectral coherence. (f) 18 Hz Eighteen and (g), and (h) 38 Hz component coherence volumes while (b) and (e) indicate the seismic amplitude with frequencies 18 and 25 Hz

respectively. The dotted lines indicate the faults. (a) shows vertical sections, (d) spectral magnitude components at 38 Hz, and (e) broadband (modified from Fangyu and Wenkai 2014)

Curvature attributes Curvature is an excellent attribute for detecting fractures and even faults. It is calculated using a very simple formula (2):

$$K = \frac{1}{R} = \frac{2\pi}{2\pi R} = \frac{dw}{dS} = \frac{d^2y/dx^2}{(1 + (dy/dx)^2)^{3/2}}, \quad (2)$$

which is also the inverse of the osculating circle with seismic horizons. According to Roberts (2001), curvature is a 2D property of a curve and describes how bent a curve is at a particular point on the curve, i.e., how much the curve deviates from a straight line at this point (Fig. 2). Therefore, positive or negative curvature values indicate anticlinal or synclinal structures respectively (Fig. 3).

Thus, there are several types of curvature attributes such as mean curvature (k_m), Gaussian curvature (k_g), maximum curvature (k_{max}), minimum curvature (K_{min}), intaglio curvature (K_d), curvature of strike (K_s), contour curvature (k_c), and most positive and negative curvatures ($K+$ and $K-$) (Fig. 4). The utility of curvatures in reservoir analysis varies according to the structures they emphasize. It is in this sense that the maximum curvature is very effective in detecting faults. Similarly, curvature measurements such as the most positive curvatures and the most negative curvatures are increasingly used by specialists because of their significant results in terms of structural analysis. Thus, the positive curvatures show anticlinal structures and raised fault blocks, while the most negative

Fig. 2 Mathematical definition of curvature. For a particular point P on a curve, the curvature can be defined in terms of the radius of curvature; R of the osculating circle with a tangent T with the curve. N is the vector normal to the curve at point P , which defines the local dip angle θ . The curvature at point P is defined as the reciprocal of the radius of curvature (modified from Roberts 2001)

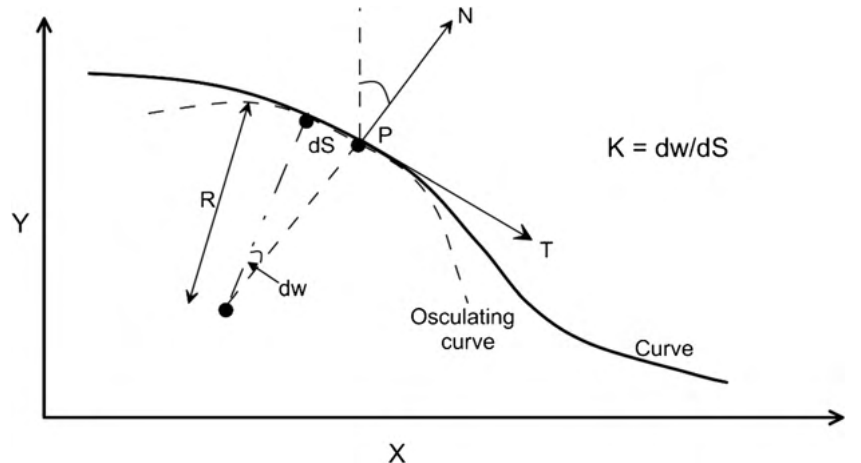


Fig. 3 Sign convention adopted for curvature values (modified from Sigismondi and Soldo 2003)



curvatures indicate the synclinal and rejected fault blocks (Chopra and Marfurt 2007). In addition to their effectiveness, the determination of traps makes it possible to highlight stratigraphic characteristics such as channels, lifts, and bars.

The first step in the reservoir study is to locate reservoir rocks by identifying structures capable of accumulating hydrocarbons. To do this, the structural attributes define the oil system by highlighting the pitfalls that constitute the essential element in the location of reservoirs. The attributes of curvature, similarity, and spectral decomposition play a large role in the location of oil deposits. Thus, studies have been conducted

to find out the location of reservoirs (AlBinHassan and Marfurt 2003; Cohen et al. 2006; Fomel 2007). Attributes such as instantaneous amplitude, spectral decomposition, curvature, and coherence enable to determine geological structures that constitute the zones where hydrocarbons are trapped. It is clear that these attributes expose structural, stratigraphic, and even mixed traps defining the hydrocarbon accumulation zones. Knowing these areas, it is necessary to make an analysis of the properties of reservoirs by seismic attributes

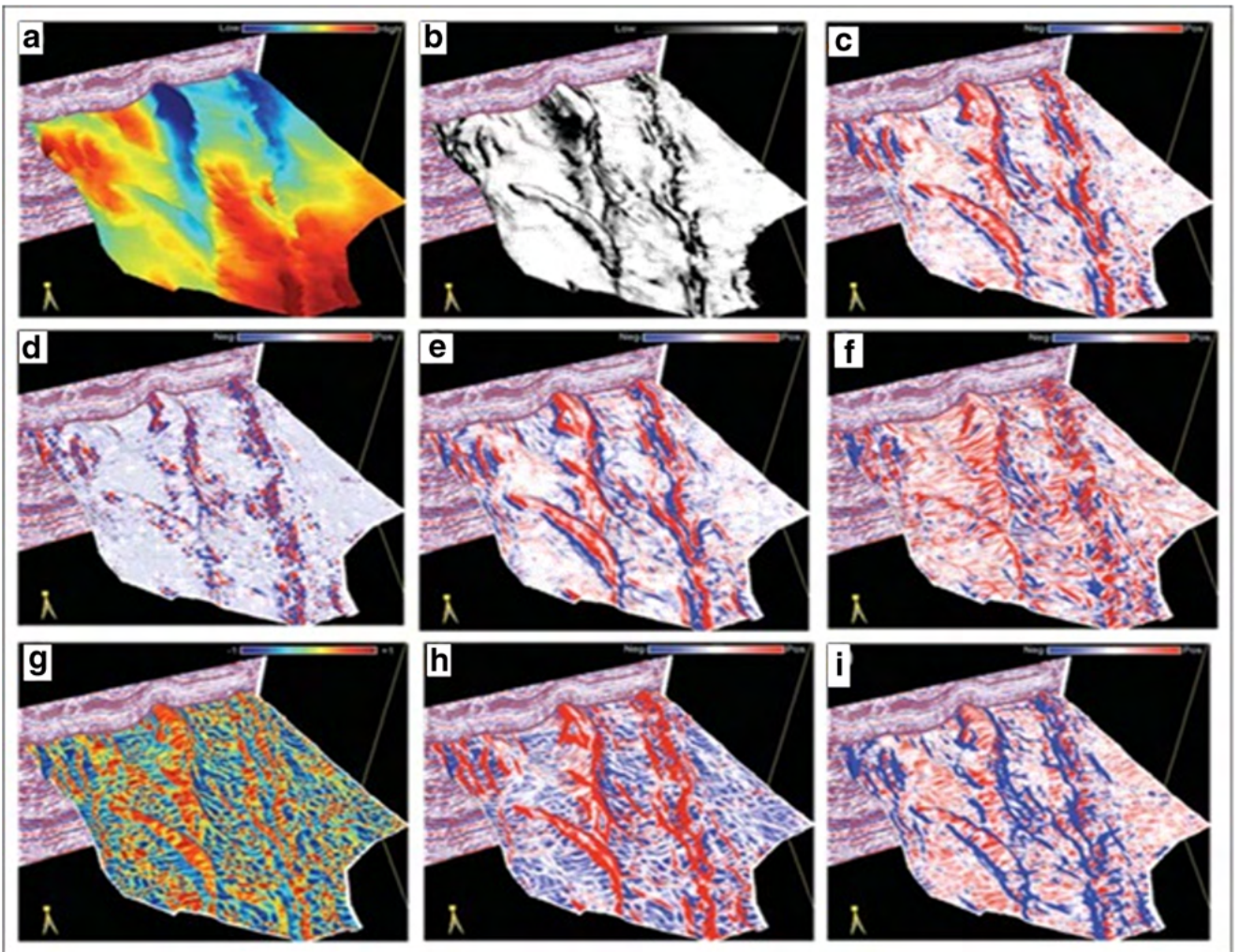


Fig. 4 The use of curvature in the detection of faults. (a) Horizon time surface. (b) Coherence. (c) Mean curvature. (d) Gaussian curvature. (e) Dip curvature. (f) Strike curvature. (g) Shape index. (h) Most positive curvature. (i) Most negative curvature (modified from Chopra and Marfurt 2007a)

Determination reservoir properties

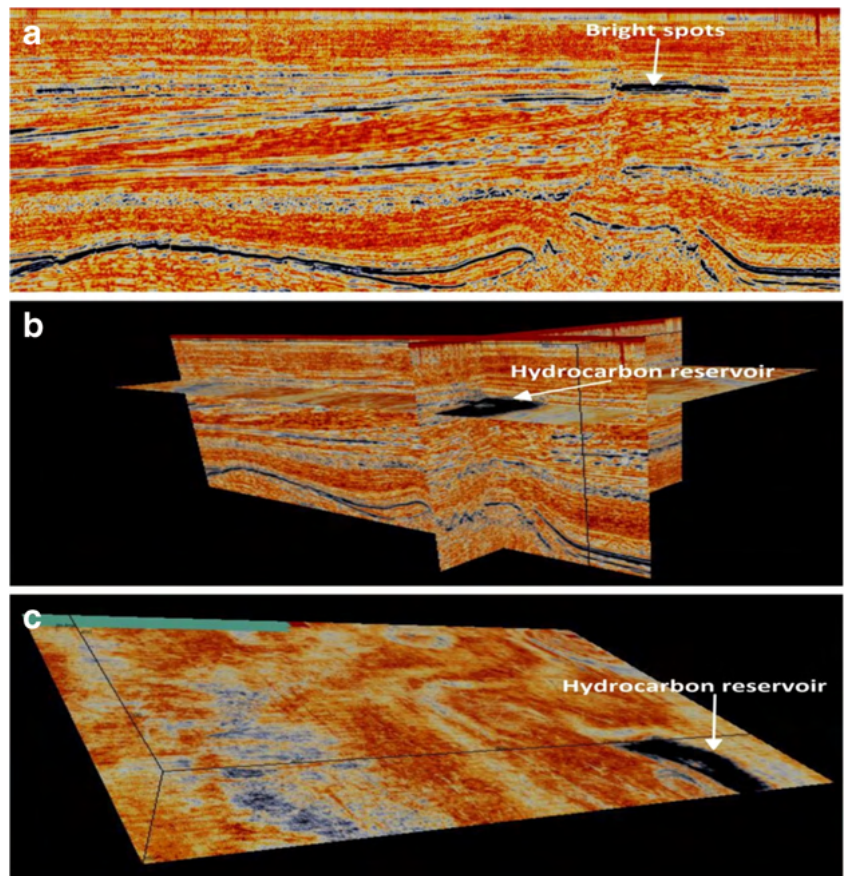
The analysis of reservoir properties is the major step in its characterization. Thus, several attributes analyze reservoir properties such as lithology, porosity, permeability, thickness, and delimitation of layer. In this case, the analysis of the reservoir parameters is performed using the seismic attributes, mainly accompanied by good data to improve the quality of results. Thus, the instantaneous amplitude, the instantaneous frequency, the spectral decomposition, the rate of variation of the envelope, the acoustic impedance, the shear impedance, the shear modulus, and the instantaneous phase are applied to extract information relating to lithology, thickness estimation, sequence delineation, porosity, and permeability. The instantaneous amplitude is widely used to accurately determine the location of reservoirs through anomalies such as bright spot, flat spot, and dim spot (Fig. 5).

Thickness and delimitation of layers

The delimitation of layers is carried out by attributes such as the instantaneous amplitude and the instantaneous phase. These two attributes highlight the sequences that exist within the reservoir. By referring to regional geology, it

becomes easy to deduce from these sequences the different layers that constitute the reservoir. After determining the different layers, it is necessary to estimate the thickness of the layers of the reservoir. To do this, the instantaneous frequency and the spectral decomposition are seismic attributes that make it possible to estimate the thickness of a layer. Sometimes, the combination of attributes provides better results in estimating thicknesses (Asim et al. 2016). It is in this sense that Zhao et al. (2008) have shown that the association of maximum amplitude and acoustic impedance illustrates the estimation of thicknesses (Fig. 6). Spectral decomposition (Castagna et al. 2003; Naseer et al. 2017; Naseer and Asim 2018) is a seismic attribute that highlights a spectrum that describes the abnormal low- and high-frequency effects responsible of seismic traces to change (Fig. 7). It delineates the reservoir and is very important in the stratigraphic analysis of reservoirs. It promotes a better understanding of reservoir heterogeneities and their implications by analyzing parameters such as seismic geomorphology, thickness estimation, lithology, and even direct hydrocarbon detection. It is widely used to estimate thickness (Liu and Marfurt 2006; Dezfoolian et al. 2013; Li et al. 2019).

Fig. 5 Illustration of bright spot detection. (a) Instantaneous amplitude of inline 230. (b) The three components of 3D seismic data which are inline, crossline, and z-slice to show the extent of the reservoir in space. (c) The hydrocarbon reservoir seen by z-slice 524 (modified from Sanda et al. 2020)



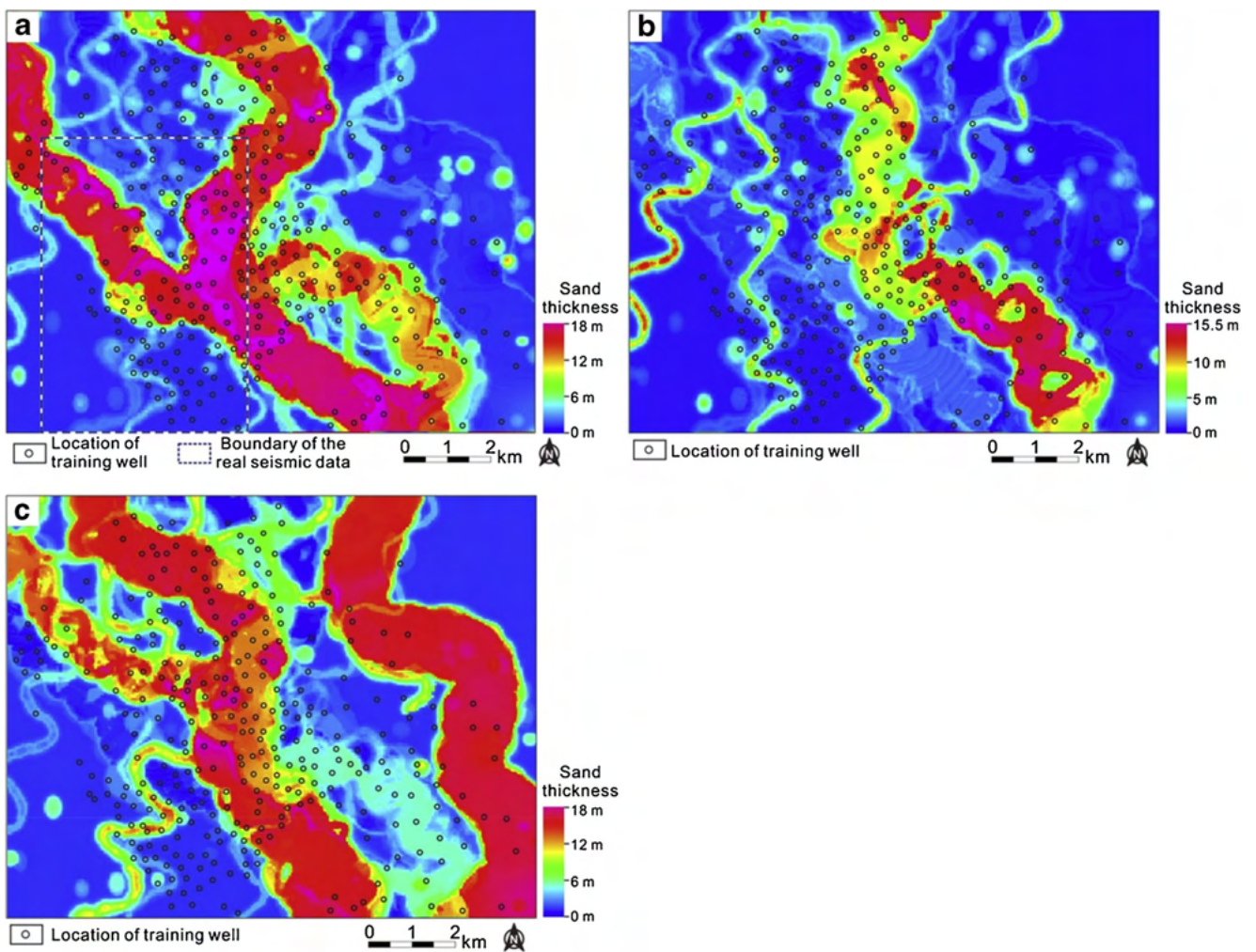


Fig. 6 Predicted sand thicknesses in three zones of 3D forward model. (a) Predicted sand thickness in zone 1 by fusing RMS amplitude of zone 1 and zone 2. (b) Predicted sand thickness in zone 2 by fusing RMS amplitude of zone 1, zone 2, and zone 3. (c) Predicted sand thickness in

zone 3 by fusing RMS amplitude of zone 2 and zone 3. The distribution of sand thickness between training wells can be regarded as cross-validated data (Li et al. 2019)

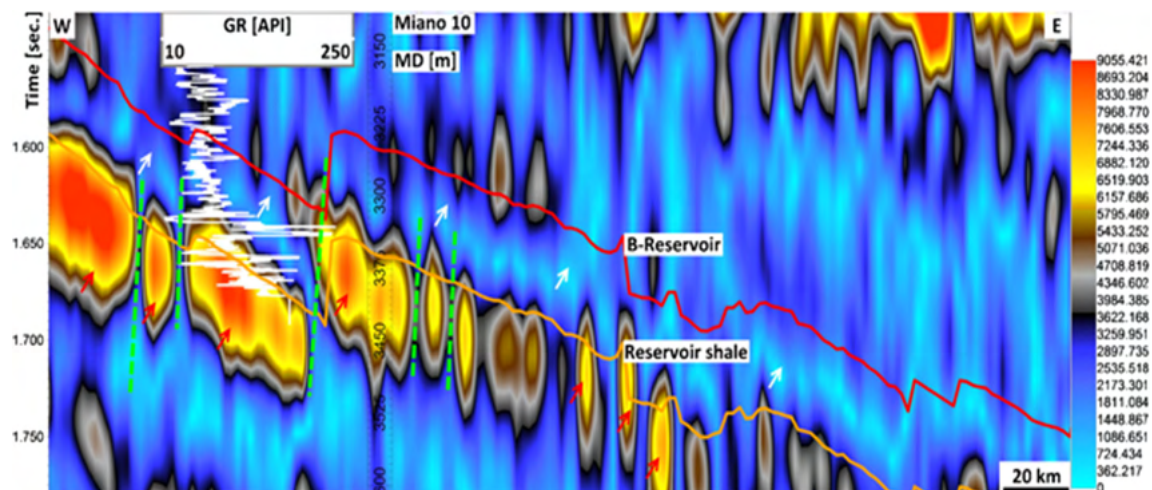


Fig. 7 Illustration of spectral decomposition, showing the discrimination of the porous reservoir (red arrows) and non-porous shales (white arrows), and the faults (green-dashed surfaces) (modified from Naseer and Asim 2018)

Fig. 8 Cross-plot of λ_ρ and μ_ρ from well logs color coded by Vshale (modified from Adeoti et al. 2017)

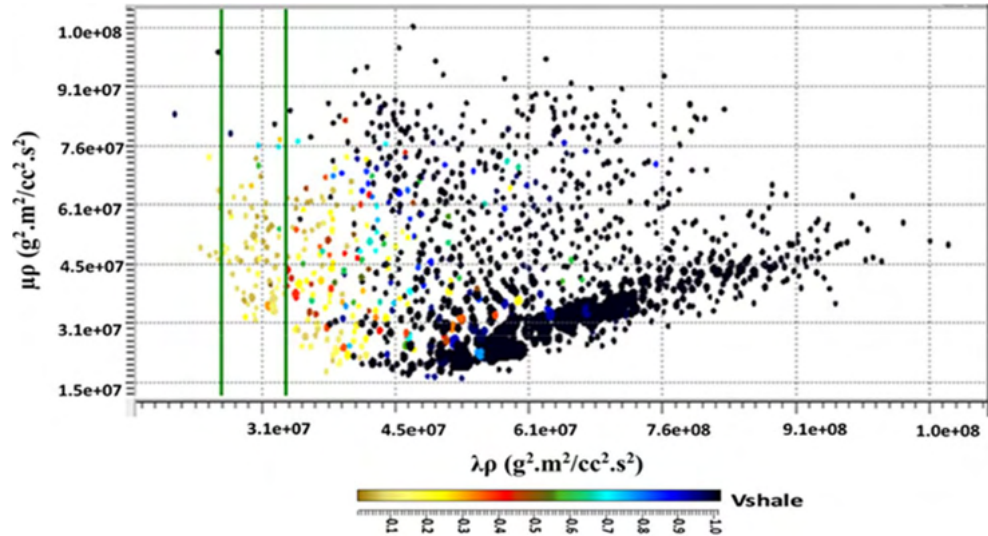
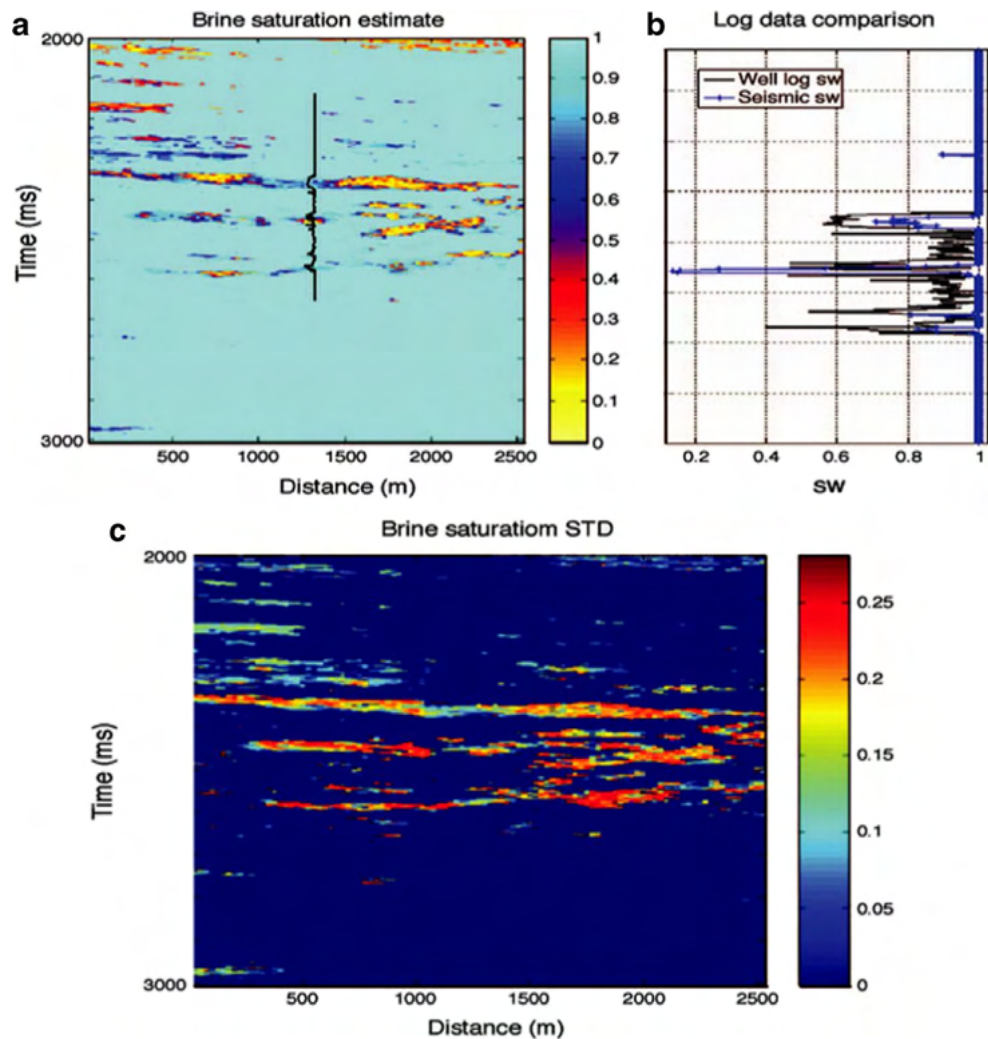


Fig. 9 (a) Estimation of brine saturation from joint inversion. (b) Comparison to well log data. (c) Saturation STD associated with the MAP estimator (modified from Bachrach 2006)



It is very important in the characterization of the reservoirs because of the information it contains on the accumulation zones (Wang and Huang 2016). The nature of the layers is a determining factor in the quality of the reservoir. The instantaneous amplitude, instantaneous frequency, the rate of variation of the envelope, and the shear modulus are attributes that determine the lithological parameters of reservoirs. The instantaneous attributes (Khan and Akhter 2016) such as amplitude, frequency, and rate of change of the envelope, through a good visualization, provide highlight of the different layers. To do this, it is important to have enough knowledge of the geology of the study area to better highlight the different layers. The extraction of the λ_ρ and μ_ρ attributes of the seismic data provides quality information on the nature of the rocks and even on the fluids of the reservoir (Fig. 8). By crossing these two attributes, the identification of different lithologies becomes easy. It is the same that the crossing of the ratio of velocities V_p/V_s by μ_ρ offers good lithological discrimination. A cross-plot of the natural log of acoustic impedance versus Poisson's ratio is often used for the identification of layer lithology (Hilterman et al. 1998).

In the study of reservoirs, porosity and permeability are the most important properties. It is clear that a good knowledge of these parameters leads to a good understanding of the quality of the study area but their determination remains difficult. Thus, the approach used by the seismic (Pride et al. 2003; Rubino et al. 2012; Naeem et al. 2015) and more precisely by the seismic attributes would allow the estimation of the porosity and the permeability through the variation of the impedance contrast obtained thanks to the reflectivity of the P and S waves on the reservoir. Thus, Bachrach (2006) uses three seismic attributes, namely acoustic impedance, shear impedance, and density, to make a joint estimate of porosity and saturation (Figs. 9 and 10). Attributes deduced from impedances such as λ_ρ , μ_ρ , and V_p/V_s are widely used to predict porosity. Similarly, the crossing of the two AVO (amplitude variation with offset) attributes, namely the gradient and the intercept, makes it possible to estimate the porosity. Thus, porosity estimation is most often done using attributes such as deterministic inversion, neural networks, and rock property prediction. As regards permeability, the use of the instantaneous frequency would make it possible to provide information on the mobility of the fluids. Using linear regression, Corbeanu et al. (2002) demonstrate that permeability is predicted from instantaneous amplitude and frequency at

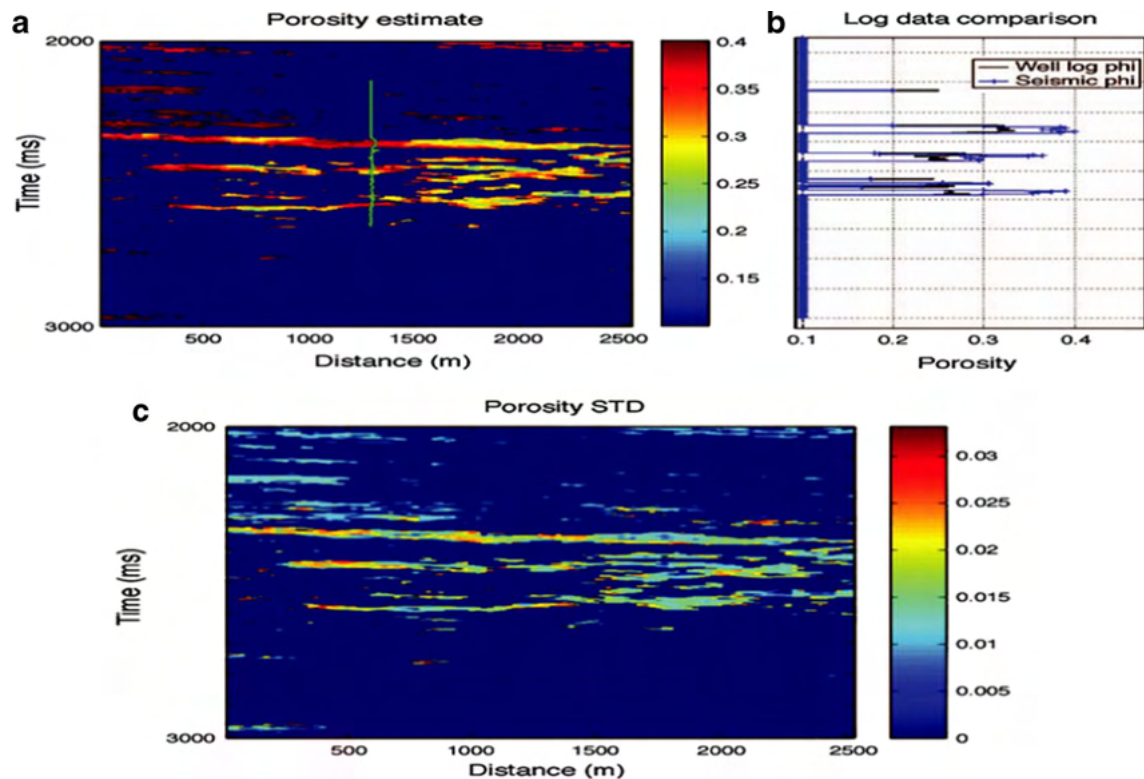
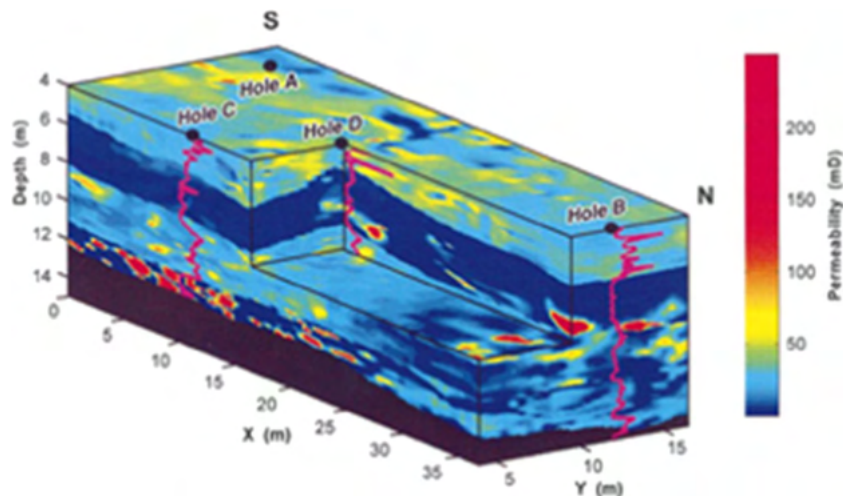


Fig. 10 a Estimation of porosity from joint inversion. b Comparison to well log data. c Porosity STD associated with the MAP estimator (modified from Bachrach 2006)

Fig. 11 Illustration of the determination of permeability by amplitude. The 3D predicted permeability cube with slices for comparing GPR amplitudes with permeabilities. The measured permeability curves are plotted in magenta at holes B, C, and D for comparison. The color bar is logarithmic to emphasize the low permeability values that dominate the cube (modified from Corbeau et al. 2002)

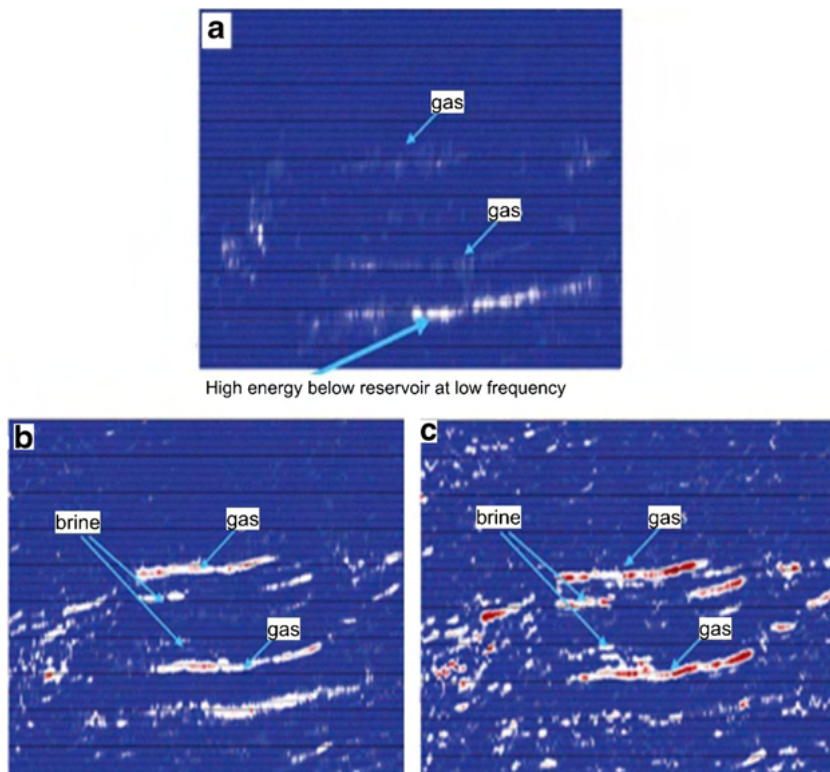


each point in the data volume (Fig. 11). Thus, Goloshubin et al. (2006) conducted a study that uses low frequencies to estimate fluid content and map highly permeable production zones in reservoirs.

Identifying areas of hydrocarbons

The identification of hydrocarbons through seismic data began a long time ago when several works have been devoted to the study of the amplitude of seismic waves in the detection of areas of hydrocarbon accumulations (O'Doherty and Anstey 1971; Sheriff 1974; Barry and Shugart 1974; Hiltebert

Fig. 12 a–c Illustration of the detection of hydrocarbons by the instantaneous frequency (modified from Castagna et al. 2003)



1975). Thus, the work of Anstey (1972, 1973a, 1973b) marks the birth of the seismic attributes which over time have become precious tools of the petroleum industry. For him, the force of reflection which is a measure of amplitude was the main attribute for the analysis of bright spots. Afterwards, researchers' interest in the notion of attributes is justified by studies carried out in this domain (Taner et al. 1979; Robertson and Fisher 1988; Taner et al. 1994; Chen and Sidney 1997; Barnes 2001; Chopra and Marfurt 2005; Zhang et al. 2015; Wang et al. 2019). In the literature, several attributes make it possible to detect hydrocarbons on a volume or a seismic section (Li and Liu 2018; Negm et al. 2020). The

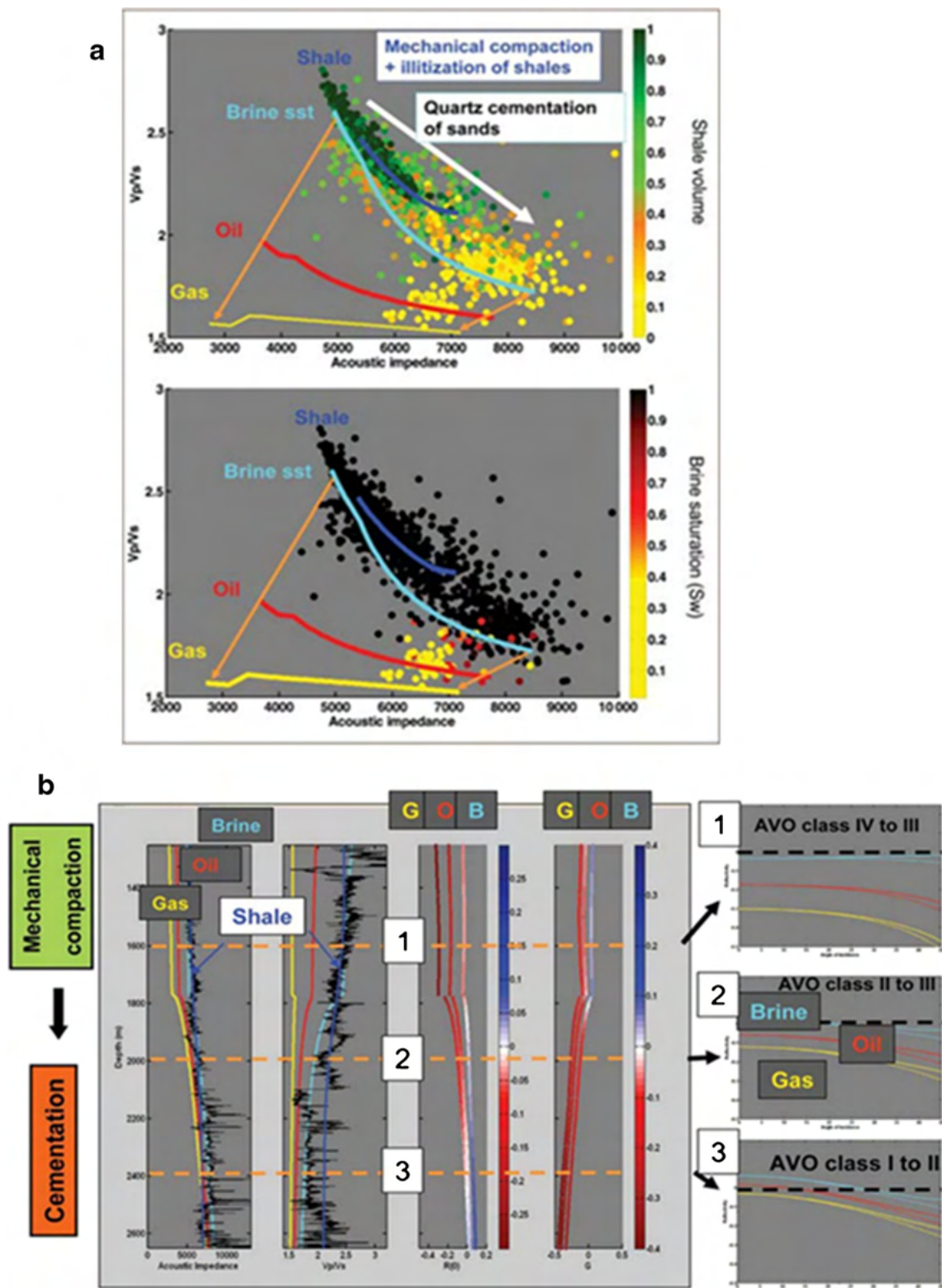


Fig. 13 (a) Illustration of the hydrocarbon detection using the cross-plotting of acoustic impedance and V_p/V_s ratio. (b) Different classes of AVO and their meanings (modified from Avseth et al. 2008)

most common are the AVO attributes, frequency attenuation, energy, sweetness, fluid contact finder, seismic feature enhancement, instantaneous amplitude, instantaneous

frequency, gas chimneys, three-term inversion, neural network classification, spectral decomposition, and the apparent polarity. Thus, the works of Tahir et al. (2016) distinguished

the gas and non-gas zones on the basis of regression analysis with attributes such as average instantaneous frequency, peak spectral frequency, average instantaneous phase, and sweetness attributes. The direct detection of hydrocarbons (bright spot, flat spot, and dim spot) by the instantaneous amplitude and the instantaneous frequency is not always effective (Fig. 12). Attenuation is defined as any decrease in amplitude of the wavefront during its propagation and is associated or not with the velocity dispersion, depending on the physical mechanism at the origin of the decrease in amplitude (Yao 2013). Also called the inverse of the quality factor, seismic attenuation is an attribute sensitive to hydrocarbon accumulation, fluid-saturated fractures, and roughness (Walls et al. 2003). The usefulness of this attribute in the characterization of reservoirs is great because the ratio between compression and shear attenuations is used as an indicator of hydrocarbons for good data. Also, the variation of frequencies is an indicator of the detection of hydrocarbons. This is how low frequencies highlight the presence of reservoirs (Castagna et al. 2003). To avoid this doubt, the use of AVO attributes appears to be a solution in the detection of hydrocarbons. The AVO attributes as demonstrated by Smith and Guidlow (1987), Rutherford and Williams (1989), Hiltermann (1990), Castagna and Smith (1994), Castagna et al. (1998), Disiena et al. (1995), Avseth et al. (2014), Xiao et al. (2019), and Farfour and Foster (2020) are involved in all stages of oil exploitation, exploration, development, and production. On the basis of the normal incidence reflection coefficient, Rutherford and Williams (1989) suggest three classes of AVO: I, II, and III (Fig. 13). The class I refers to sands characterized by high impedance compared to overlying shales with a positive intercept (a) and a negative gradient (b). The class II is composed of sands with low normal incidence reflectivity and the class III is represented by sands with lower impedance than overlying shales and increasing amplitude of reflection with offset (intercept (a) and gradient (b) are negative). The class IV is considered when the reflection coefficients change more and more from negative sign to positive sign with an increasing offset (Castagna et al. 1998). Hiltermann et al. (2000) demonstrated that class III and IV anomalies are better at recognizing amplitude bright spots on seismic data, while class II anomalies are needed simply to identify potential reservoirs. To do this, petrophysical parameters such as impedance, P and S wave velocities, energy conversion elasticity modules, and wave conversion are extracted to estimate the fluid content of the diagrams of these parameters. The use of these attributes for hydrocarbon identification and recognition is very effective. The detection of hydrocarbons by AVO attributes is a topic of interest to specialists of this field (e.g., Smith and Guidlow 1987; Fatti et al. 1994; Veeken and Rauch-Davies 2006).

Discussion

The characterization of reservoirs by seismic attributes is an interesting technique that provides quality results which allows firstly to locate the accumulation zones, then to analyze the characteristics of reservoirs, and finally to determine the distribution of hydrocarbons. The use of seismic attributes to characterize reservoirs provides information on reservoir characteristics such as porosity, thickness, lithology, direct hydrocarbon indicators (bright spot, flat spot, and dim spot), and traps (faults, anticlines, synclines, and salt domes). It should be emphasized that this approach can be improved by combining it with other methods such as log analysis, seismic sampling, neural networks, and statistical analysis. However, the characterization of the reservoirs becomes a multidisciplinary study because it integrates geostatistics, geophysics, petrophysics, seismic coring, and geology.

By associating seismic attributes with good logs, all petrophysical properties (porosity, permeability, water saturation, clay volume, and/or shale volume) are taken into account. This approach provides precise information on the property values of reservoirs (Sofolabo and Onyia 2017). It becomes very easy to interpolate the well data with the seismic attributes to know the distribution of the properties over the whole extent of the reservoir. In addition, gamma ray and resistivity logs provide valuable information on reservoir characteristics (Eshimokhai and Akhievbulu 2012). The integration of good logs with seismic data provides more accurate seismic attributes in the analysis of hydrocarbon reservoir characteristics.

The statistical approach to reservoir characterization is an important tool in the analysis of the petrophysical properties of reservoirs in that it allows the values of each characteristic to be calculated. As this approach provides local information, it is important to use the seismic attributes derived from the integration of statistical data with seismic data. This leads to a joint analysis of several attributes called multiattribute (MA). To do this, the multiattribute analysis highlights three approaches, namely the co-kriging technique, the neural networks (Aminzadeha et al. 2000), and the covariance matrix. These three statistical methods make it possible to predict a property by using several attributes in order to improve the results obtained with an attribute.

Conclusion

By revisiting the work related to the application of seismic attributes in the characterization of reservoirs, it appears that each attribute gives a particular function and explanation for the understanding of geological structures and hydrocarbon potentials. This contribution of the attributes is manifested by the quality of the results obtained and which has been improved thanks to the combination of several attributes in order to

extract the best information. The variation of inclination and azimuth provides a rapid and reliable estimation of the convergence, divergence, parallelism, or disorder of the reflectors, which are the basis of stratigraphic or structural interpretations. So, the variation of the seismic trace is the consequence of the change of structures, stratigraphy, porosity, permeability, and presence of hydrocarbons in geological layers. Despite the multifunctionality, each attribute is very effective in a specific field of reservoir characterization. It is therefore important to specify that the instantaneous amplitude and frequency are good indicators of the presence of gas. The instantaneous phase is very effective to determine the configuration of the continuity of lateral reflector and bedding. The spectral decomposition is important in estimating bed thickness and finally coherence while curvature is the best attribute for detecting faults, fractures, and lateral stratigraphic discontinuities. Seismic attributes play a very important role in identifying potential hydrocarbon zones. Sometimes, these attributes directly detect hydrocarbons and refer to direct indicator of hydrocarbons. Among these attributes, the AVO attribute explains the lithology as well as the presence of the fluids on a seismic volume (Welsh et al. 2008). This analysis shows that the attributes most commonly used in the detection of hydrocarbons are those related to the amplitudes and frequencies of seismic traces.

Single attribute analysis uses a linear or non-linear relationship between a seismic attribute and a property using a single trace parameter (Guerrero et al. 2010). Generally, it is important to combine several attributes in order to improve the quality of the results obtained during the analysis of reservoirs, hence the emergence of the notion of multiattributes which integrates within it all the geostatic methods that use the attributes to characterize the reservoirs. Thus, among these attributes, there is the co-kriging, the neural networks, and the covariance matrix.

Compliance with Ethical Standards

Conflict of interest The authors declare that they have no competing interests.

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